SECTION A: CONCEPT APPLICATION

Q1. Identify the string methods and their execution order needed to remove extra spaces and fix inconsistent casing in a product name.

Answer: The strip() method removes extra spaces from the beginning and end of a string. The title() method converts the first letter of each word into uppercase and the remaining letters into lowercase. The correct execution order is strip() followed by title().

Q2. Explain why manual CSV loading often treats numeric columns as strings and identify the function used to convert them to floats.

Answer: CSV files store data in a text-based format. Therefore, when values are manually read from a CSV file, numeric values may be treated as strings. We use the float() function to convert string values into floating-point numbers so that mathematical calculations can be performed.

Q3. Determine the best data structure for mapping product names to total sales and explain the drawback of using a list instead.

Answer: A dictionary is the best data structure for mapping product names to their total sales because it stores information in key-value pairs. The product name is used as the key, and total sales are stored as the value. A list is less suitable because it stores values using index positions, making it difficult to identify and manage the relationship between product names and sales values.

Q4. Compare using a manual for loop versus the sum() function, identifying a scenario where the loop is the only viable option.

Answer: The sum() function is used to calculate the total of numerical values. It is simple and requires less code. A manual for loop adds values one by one and provides more

control over calculations. For example, if we want to calculate total sales only for transactions greater than 2,000, we can use a for loop with an if condition.

Q5. Describe how to implement type checking or error handling to prevent a TypeError when comparing a potential string variable to a number.

Answer: We can use the isinstance() function to check whether a variable contains an integer or float before comparing it with a number. We can also use float() to convert string values into numbers. Additionally, try-except error handling can catch errors such as TypeError and ValueError, preventing the program from stopping unexpectedly.

Q6. Explain the benefits of modularizing a data pipeline into separate functions and the risks of using a single large code block in analytics.

Answer: Modularizing a data pipeline means dividing a large program into smaller functions, where each function performs a specific task such as loading data, cleaning data, converting data types and calculating sales. It makes the program easier to understand, maintain, reuse and debug. Using one large code block makes the program difficult to manage, troubleshoot and update.